MATH3303 – Lecture 31

Irreducible $\iff$ Prime in a PID; Unique Factorisation Domains

Review Notes

Reference: Lee, Abstract Algebra*, Sections 10.3–10.4*

Overview

This lecture closes the irreducible-vs-prime gap inside a PID, and then introduces *unique factorisation domains* (UFDs).

- **Completion of the irreducible $\iff$ maximal lemma** in a PID
- **Theorem (in a PID):** irreducible $\iff$ prime
- **Corollary (in a PID):** nonzero proper ideal is prime $\iff$ maximal
- **Definition of a UFD**
- **Examples:** $\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Z}[x]$, $K[x, y]$; the Heegner numbers; cyclotomic integers $\mathbb{Z}[\zeta_p]$ and the historical context of Fermat's Last Theorem
- **Non-example:** $\mathbb{Z}[\sqrt{-5}]$ is not a UFD
- **Theorem (Claim 1):** every PID has factorisation into irreducibles

Connection to Lecture 30. Lecture 30 proved the ($\Rightarrow$) direction of the irreducible-iff-maximal lemma in a PID. We now complete the ($\Leftarrow$) direction, then chain the lemma with results from Lectures 25–29 to conclude that prime, irreducible, and maximal coincide (for nonzero principal ideals) in a PID – and use this to launch the UFD theory.

1 Completing the Irreducible $\iff$ (r) Maximal Lemma

Lemma 1.1: Irreducible $\iff$ (r) maximal in a PID (from lecture)

Let R be a PID and $r \in R$. Then

$$r \text{ is irreducible} \iff (r) \text{ is a maximal ideal.}$$

Proof. *(⇒)* See Lecture 30, Lemma 5.1.

(⇐) Suppose (r) is maximal. Then r is not a unit (otherwise $(r) = R$, contradicting maximality) and $r \neq 0$ (otherwise $(r) = (0)$ is not maximal in a nontrivial PID).

Suppose $r = ab$ for some $a, b \in R$. Then $(r) \subseteq (a) \subseteq R$. Since (r) is maximal, either $(a) = R$ or $(r) = (a)$.

- If $(a) = R$, then $1 \in (a)$, so $1 = ac$ for some $c \in R$, i.e. a is a unit.
- If $(r) = (a)$, then $a \in (r)$, so $a = rx$ for some $x \in R$. Hence

$$r = ab = rxb \implies r(1 - xb) = 0.$$

  Since $r \neq 0$ and R is an integral domain, $1 - xb = 0$, i.e. $xb = 1$. So b is a unit.

Either way, a or b is a unit. Hence r is irreducible. □

2 Irreducible $\iff$ Prime in a PID

Theorem 2.1: Irreducible $\iff$ prime in a PID (from lecture; cf. Lee, Thm. 10.11)

Let R be a PID and $r \in R$. Then

$$r \text{ is irreducible} \iff r \text{ is prime.}$$

Proof. *(⇒)* Suppose r is irreducible. By Lemma 1.1, (r) is maximal, hence prime (Lecture 26, Proposition 3.1). By Lecture 30, Lemma 1.2, r is a prime element.

(⇐) This is Theorem 2.2 of Lecture 30 (every prime element in an integral domain is irreducible). □

Corollary 2.1: Prime $\iff$ maximal for principal ideals in a PID (from lecture)

Let R be a PID. A nonzero proper ideal of R is prime if and only if it is maximal.

Proof. "⇐" is Proposition 3.1 of Lecture 26 (every maximal ideal is prime).

"⇒": Let $I = (r) \trianglelefteq R$ be nonzero and prime. Then

$$\begin{aligned}
(r) \text{ prime} &\iff r \text{ prime} && \text{(Lecture 30, Lemma 1.2)}\\
&\iff r \text{ irreducible} && \text{(Theorem 2.1)}\\
&\iff (r) \text{ maximal} && \text{(Lemma 1.1).}
\end{aligned}$$

□

3 Unique Factorisation Domains

Definition 3.1: Unique factorisation domain (UFD) (from lecture; cf. Lee, Def. 10.12)

A *unique factorisation domain* (UFD) is an integral domain R such that

(1) every nonunit $r \in R \setminus \{0\}$ has a factorisation

$$r = p_1 p_2 \cdots p_\ell$$

into irreducibles;

(2) if $r = q_1 q_2 \cdots q_k$ is another such factorisation, then $k = \ell$ and after permutation, p_i is an associate of q_i for each i.

Example 3.1: Examples of UFDs (from lecture; cf. Lee, Examples 10.16, 10.17)

(1) $\mathbb{Z}$ is a UFD. E.g. $12 = 3 \cdot 2 \cdot 2 = 3 \cdot (-2) \cdot (-2)$ – two factorisations differing only by units.

(2) $\mathbb{Z}[i]$ is a UFD. E.g. $5 = (1 + 2i)(1 - 2i) = (-2 + i)(2 + i)$.

(3) $\mathbb{Z}[x]$ is a UFD but *not* a PID. Similarly $K[x, y]$ (with K a field) is a UFD but not a PID.

4 $\mathbb{Z}[\sqrt{-5}]$ is NOT a UFD

Example 4.1: $\mathbb{Z}[\sqrt{-5}]$ is not a UFD (from lecture; cf. Lee, Example 10.19)

Recall (Lecture 30) that 3 is irreducible in $\mathbb{Z}[\sqrt{-5}]$. By a similar argument, 2 and $1 \pm \sqrt{-5}$ are also irreducible.

Direct proof of non-uniqueness. We have two factorisations of 6:

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

Each of 2, 3, $1 + \sqrt{-5}$, $1 - \sqrt{-5}$ is irreducible. Using the norm N:

- $N(2) = 4$, $N(1 \pm \sqrt{-5}) = 1 + 5 = 6$.
- The units are ± 1 (Lecture 30, Example 3.2(2)).
- So 2 is not an associate of $1 \pm \sqrt{-5}$ (their norms differ).

Hence the two factorisations differ in more than just associates and order. $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

Remark 4.1: Heegner numbers (from lecture)

$\mathbb{Z}[\sqrt{-d}]$ is a UFD if and only if d is one of the nine *Heegner numbers*:

$$1,\ 2,\ 3,\ 7,\ 11,\ 19,\ 43,\ 67,\ 163.$$

Example 4.2: Cyclotomic integers and Fermat's Last Theorem (from lecture)

Let $\zeta_p = \exp(2\pi i/p)$ be a primitive p-th root of unity (with p prime). The ring

$$\mathbb{Z}[\zeta_p]$$

is called the ring of *cyclotomic integers*.

Fact. $\mathbb{Z}[\zeta_p]$ is a UFD $\iff p \leq 19$.

Historical note. A naive approach to Fermat's Last Theorem (Lamé, 1847) starts from

$$x^p + y^p = (x+y)(x+\zeta_p y)\cdots(x+\zeta_p^{p-1}y)$$

working in $\mathbb{Z}[\zeta_p]$, which was mistakenly assumed to be a UFD. Its failure to be a UFD for $p \geq 23$ is what made the naive approach fail and motivated Kummer's theory of ideal numbers.

5 Every PID is a UFD: Existence of Factorisation

Theorem 5.1: Every PID is a UFD (from lecture; cf. Lee, Thm. 10.12)

Every PID is a UFD.

The proof has two claims, mirroring the two parts of the UFD definition. We prove Claim 1 (existence) below.

Proof of Theorem 5.1, Claim 1: existence of factorisation into irreducibles. Let R be a PID.

Claim 1. Every nonzero nonunit element of R can be written as a product of one or more irreducibles.

Take any nonzero nonunit $a \in R$.

- If a is irreducible, done.
- If not, then a is reducible, so $a = a_1 b_1$ where a_1, b_1 are both nonunits.

Compare ideals. $a = a_1 b_1 \in (a_1)$, so $(a) \subseteq (a_1)$. In fact this inclusion is strict.

Indeed, suppose for contradiction $(a) = (a_1)$. Then $a_1 = ar$ for some $r \in R$, so

$$a = a_1 b_1 = a r b_1.$$

Cancelling a (using $a \neq 0$ and R an integral domain): $1 = r b_1$. So b_1 is a unit, contradicting our assumption.

Hence $(a) \subsetneq (a_1)$.

Iterate the process for any remaining reducible factors. We would get a strictly increasing chain

$$(a) \subsetneq (a_1) \subsetneq (a_2) \subsetneq \cdots$$

By the ACC for PIDs (Lecture 29, Theorem 3.1), this chain stabilises after finitely many steps. So eventually every remaining factor is irreducible, giving

$$a = p_1 p_2 \cdots p_r,$$

each p_i irreducible. □

Remark 5.1: Looking ahead (from lecture)

Claim 2 (uniqueness) – that any two such factorisations match up to order and associates – is completed in the next lecture. The key tool will be Theorem 2.1: in a PID, every irreducible is prime, so we can use the prime divisibility property to peel off matching factors one at a time.

Checklist of Skills

- ☐ Prove that in a PID, (r) maximal $\Rightarrow$ r irreducible (completing the lemma from Lecture 30).
- ☐ State and prove the equivalence: in a PID, r is irreducible iff r is prime.
- ☐ Deduce the corollary: in a PID, a nonzero proper ideal is prime iff it is maximal.
- ☐ State the definition of a unique factorisation domain.
- ☐ Cite the standard examples ($\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Z}[x]$, $K[x,y]$) and the non-example $\mathbb{Z}[\sqrt{-5}]$.
- ☐ Prove directly that $\mathbb{Z}[\sqrt{-5}]$ is not a UFD using $6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$.
- ☐ State the Heegner classification of imaginary quadratic UFDs.
- ☐ Outline the historical role of $\mathbb{Z}[\zeta_p]$ in attempts at Fermat's Last Theorem.
- ☐ Use the ACC to prove that every nonzero nonunit in a PID has a factorisation into irreducibles.